Sparklint

a Tool for Identifying and Tuning Inefficient Spark Jobs Across Your Cluster

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Why Sparklint?

- A successful Spark cluster grows rapidly
- Capacity and capability mismatches arise
- Leads to resource contention
- Tuning process is non-trivial
- Current UI operational in focus

We wanted to understand application efficiency

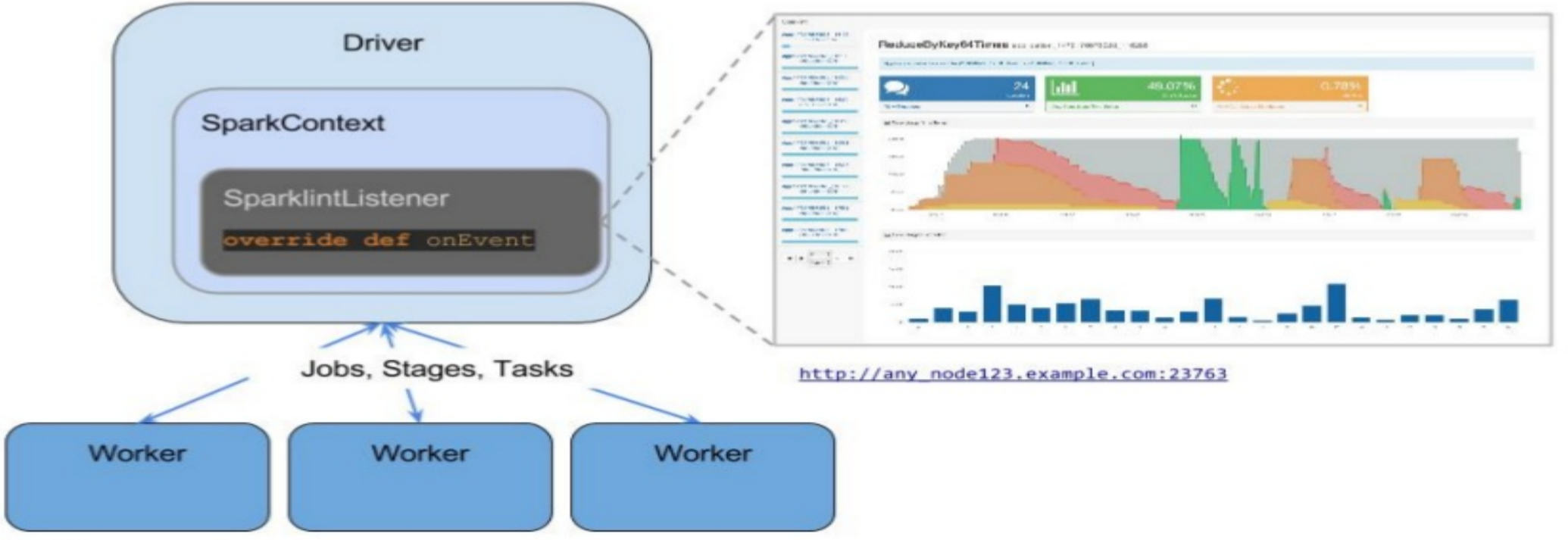
Sparklint provides:

- Live view of batch & streaming application stats

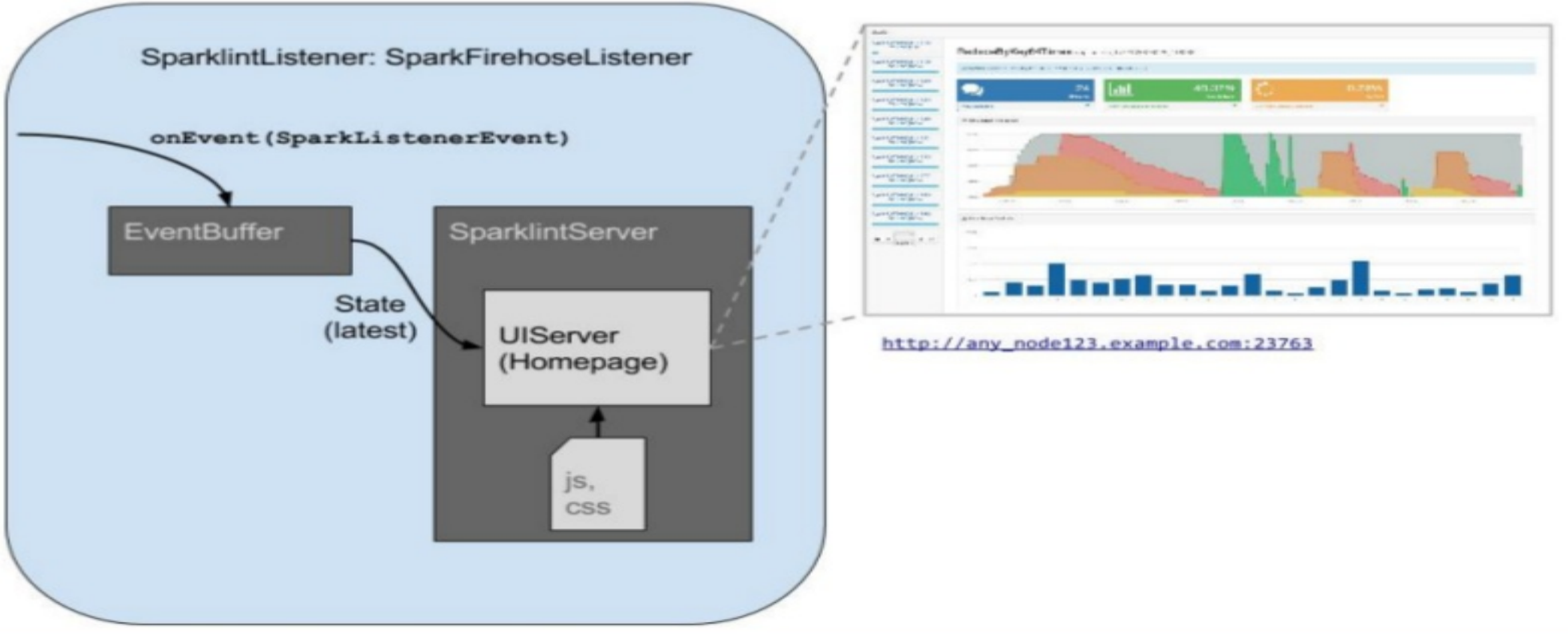
  or

- Event by event analysis of historical event logs
- Stats and graphs for:
  - Idle time
  - Core usage
  - Task locality

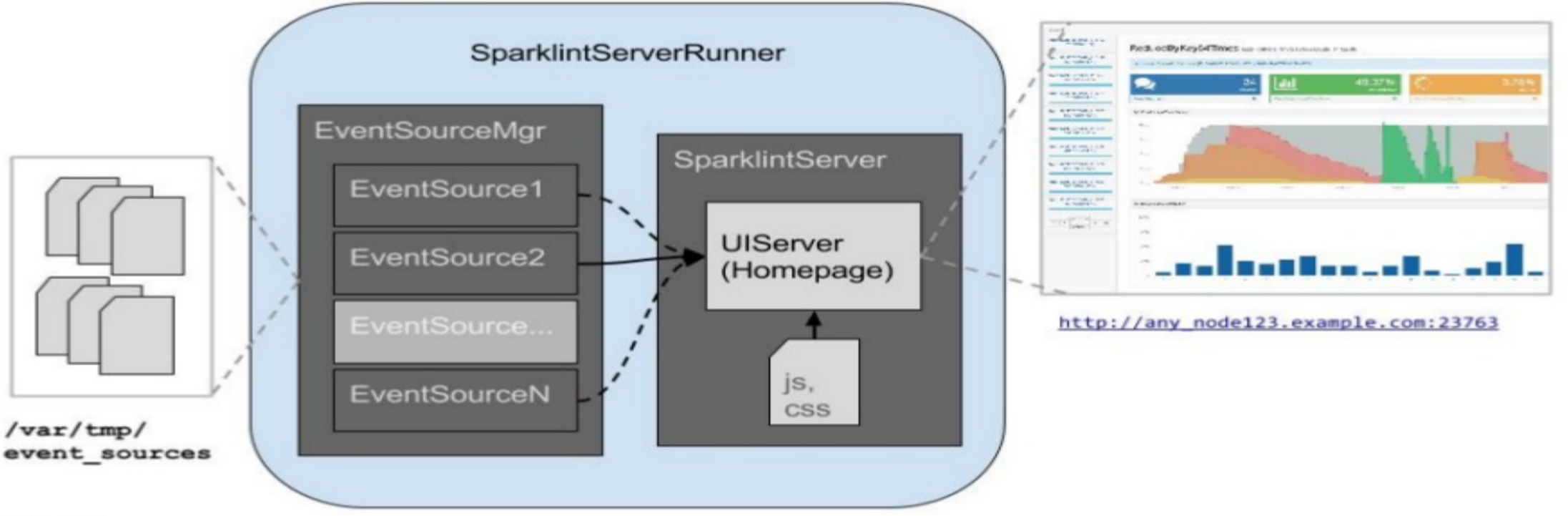
Sparklint Listener:



Sparklint Listener:



Sparklint Server:



Demo…

- Simulated workload analyzing site access logs:
  - read text file as JSON
  - convert to Record(ip, verb, status, time)
  - countByIp, countByStatus, countByVerb

Sample_16cores application_1472176676028_544163

Job took 10m7s to finish

Application finished in 10 minutes (2016-10-20T01:48:30.390Z -> 2016-10-20T01:58:37.632Z)

Already pretty good distribution; low idle time indicates good worker usage, minimal driver node interaction in job

4 Executors

View Executors

48.80% Core Utilization

View Core Usage Time Series

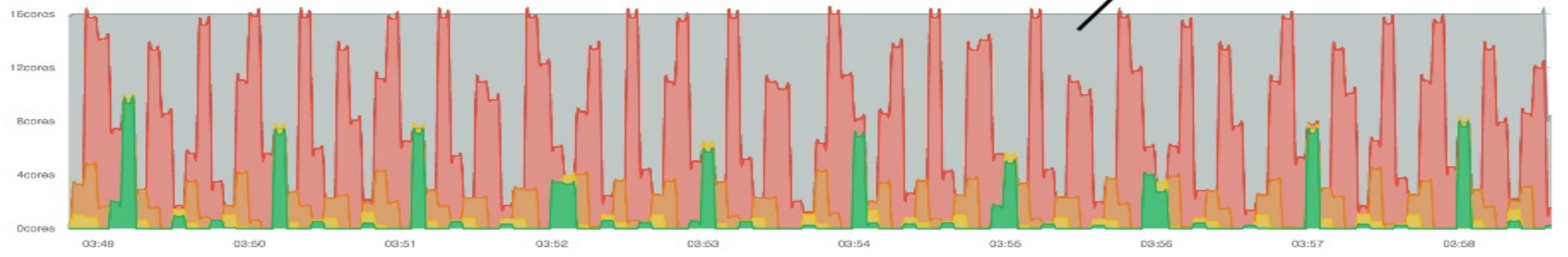
0% Idle Time

View Core Usage Distribution

But overall utilization is low

Which is reflected in the common occurrence of the IDLE state (unused cores)

Core Usage Time Series



Sample_8cores application_1472176676028_544172

Job took 15m14s to finish

Core usage increased, job is more efficient, execution time increased, but the app is not cpu bound

Application finished in 15 minutes (2016-10-20T01:49:17.941Z -> 2016-10-20T02:04:31.009Z)

2 Executors

View Executors

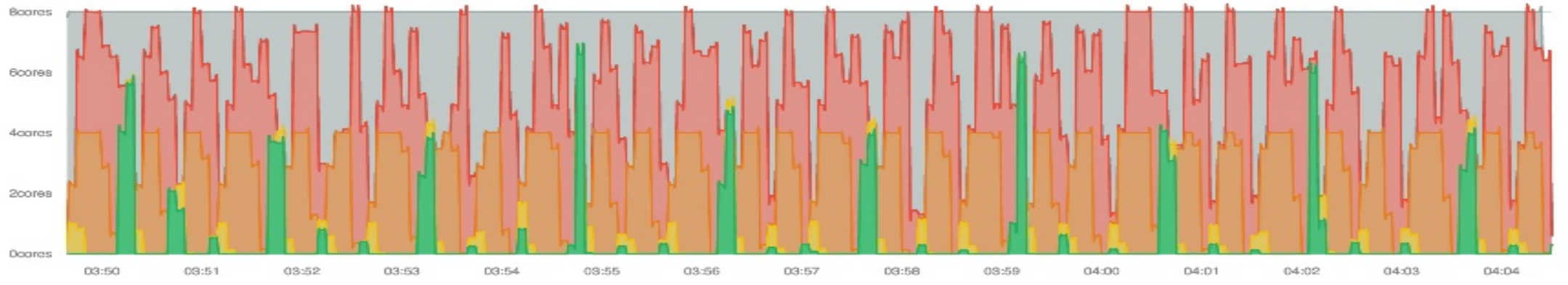
67.19% Core Utilization

View Core Usage Time Series

0% Idle Time

View Core Usage Distribution

Core Usage Time Series



Sample_32cores application_1472176676028_544282

Job took 9m24s to finish

Core utilization decreased proportionally, trading execution time for efficiency

Application finished in 9 minutes (2016-10-20T02:03:31.720Z -> 2016-10-20T02:12:55.935Z)

8 Executors

View Executors

37.77% Core Utilization

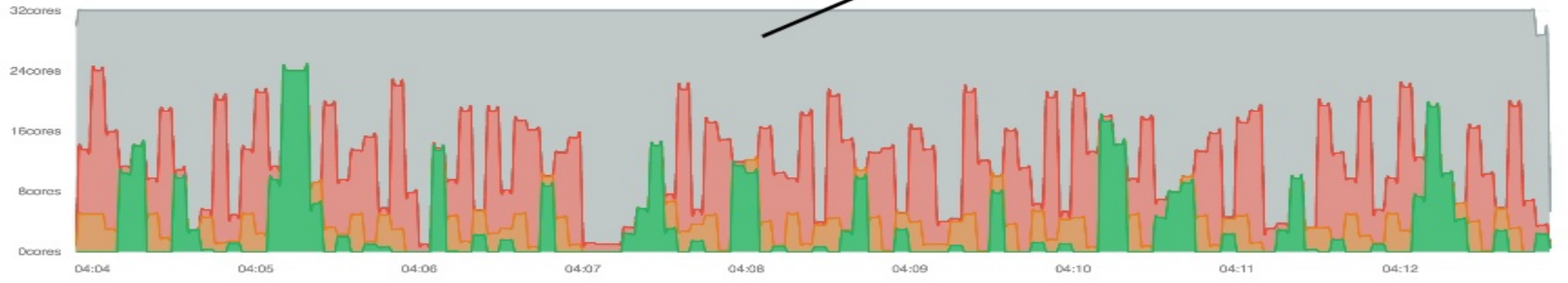
View Core Usage Time Series

0% Idle Time

View Core Usage Distribution

Lots of IDLE state shows we are over allocating resources

Core Usage Time Series



Sample_DynamicAllocation application_1472176676028_544292

Job took 11m34s to finish

Core utilization remains low, the config settings are not right for this workload.

Application finished in 12 minutes (2016-10-20T02:05:58.773Z -> 2016-10-20T02:17:32.291Z)

24 Executors

View Executors

41.02% Core Utilization

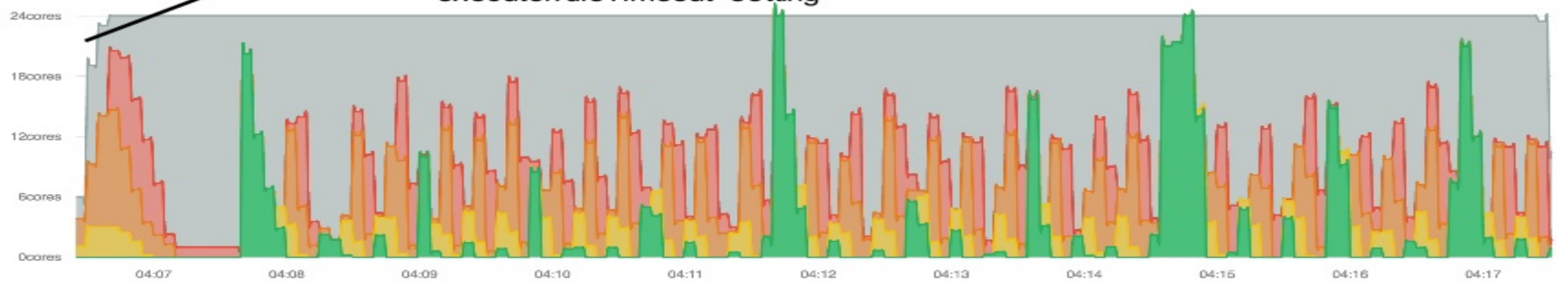
View Core Usage Time Series

0% Idle Time

View Core Usage Distribution

Core Usage Time Series

Dynamic allocation only effective at app start due to long executorIdleTimeout setting



Sample_DynamicAllocation_10s application_1472176676028_550039

Job took 33m5s to finish

Application finished in 33 minutes (2016-10-20T15:47:39.867Z -> 2016-10-20T16:20:52.851Z)

Core utilization is up, but execution time is up dramatically due to reclaiming resources before each short running task.

Executor churn!

940 Executors

View Executors

62.54% Core Utilization

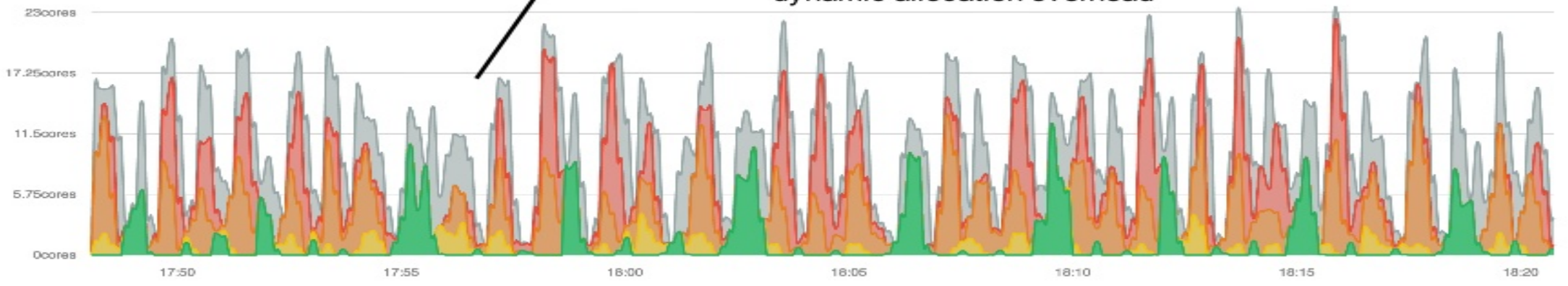
View Core Usage Time Series

0% Idle Time

View Core Usage Distribution

IDLE state is reduced to a minimum, looks efficient, but execution is much slower due to dynamic allocation overhead

Core Usage Time Series



Sample_16cores_parallel application_1472176676028_555221

Job took 7m34s to finish

Core utilization way up, with lower execution time

Application finished in 8 minutes (2016-10-21T03:40:57.744Z -> 2016-10-21T03:48:29.451Z)

4 Executors

View Executors

80.60% Core Utilization

View Core Usage Time Series

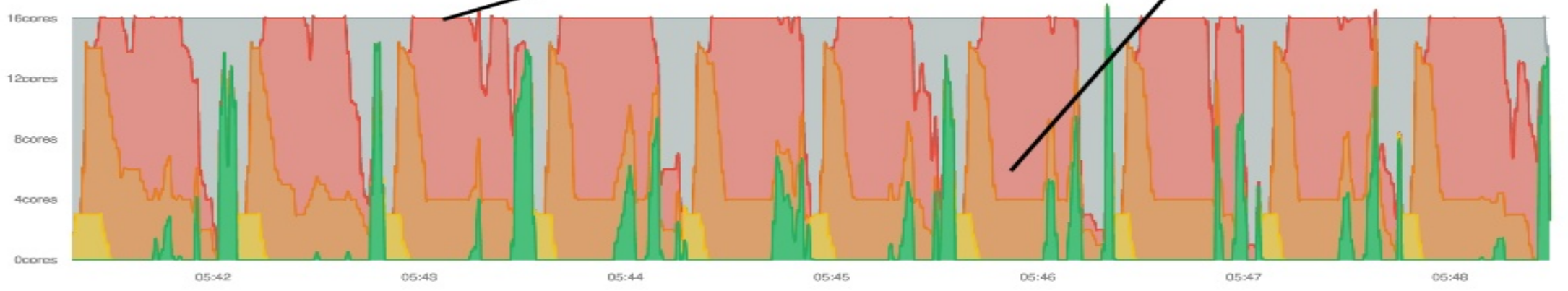
0% Idle Time

View Core Usage Distribution

Flat tops show we are becoming CPU bound

Parallel execution is clearly visible in overlapping stages

Core Usage Time Series



Sample_32cores_parallel application_1472176676028_555209

Job took 5m6s to finish

Core utilization decreases, trading execution time for efficiency again here

Application finished in 5 minutes (2016-10-21T03:39:10.199Z -> 2016-10-21T03:44:16.894Z)

8 Executors

View Executors

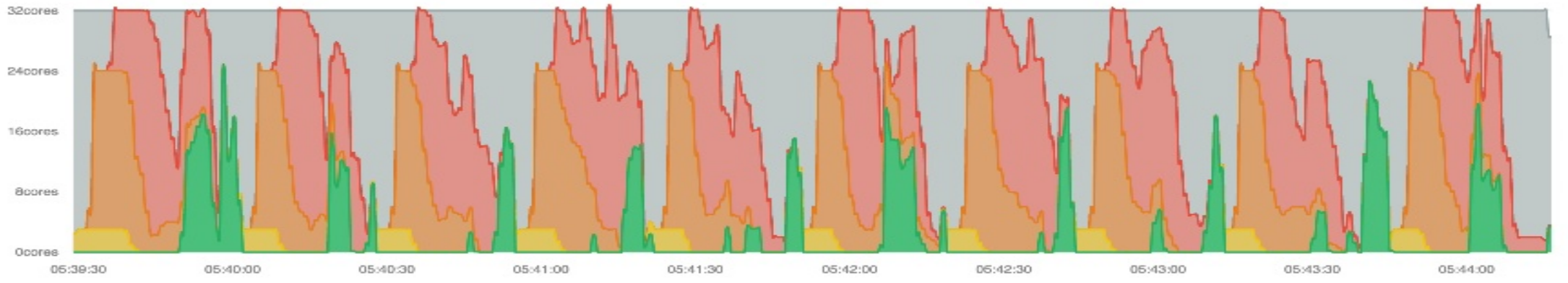
60.74% Core Utilization

View Core Usage Time Series

0% Idle Time

View Core Usage Distribution

Core Usage Time Series



junoHourly-2016-10-05:21 application_1472176676028_407508

Application finished in 43 minutes (2016-10-05T22:11:39.283Z -> 2016-10-05T22:54:09.709Z)

Thanks to dynamic allocation the utilization is high despite being a bi-modal application

58 Executors

View Executors

82.02% Core Utilization

View Core Usage Time Series

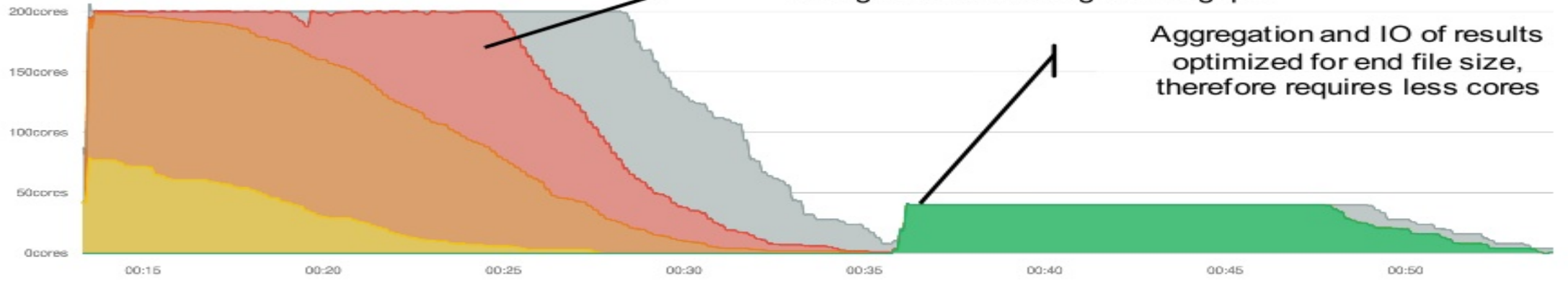
0.39% Idle Time

View Core Usage Distribution

Core Usage Time Series

Data loading and mapping requires a large core count to get throughput

Aggregation and IO of results optimized for end file size, therefore requires less cores



Future Features:

- Increased job & stage detail in UI
- History Server event sources
- Inline recommendations
- Auto-tuning
- Streaming stage parameter delegation

The Credit:

- Lead developer is Robert Xue
- https://github.com/roboxue
- SDE @ Groupon

Contribute!

Sparklint is OSS:

https://github.com/groupon/sparklint

THANK YOU.

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